SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

***CO2:** Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Steffi.I (192524480) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Steffi.I

20+28+20+18+8

94

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Local search / optimization

Given :

$R1 = 120$ km
$R2 = 95$ km
$R3 = 105$ km
$R4 = 85$ km

Current solution $= R1$

Local search improvement :

Start from R1 (120 km)

$R2 = 95$ km
$R3 = 105$ km
$R4 = 85$ km

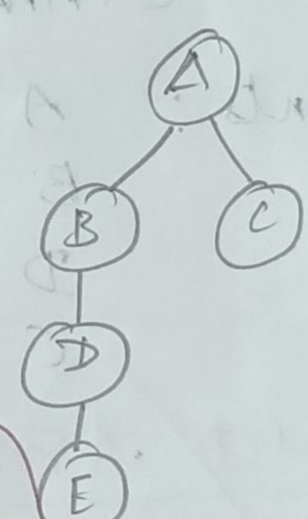
Move step by step

$R1 \rightarrow R2$ (95 km)
$R2 \rightarrow R4$ (85 km)

Final solution : $R4 = 85$ km (optimal Route)

2) Online search Agent

$A \rightarrow B$
$A \rightarrow C$



Step-by-step

1) Start at A
→ Known : B, C

2) Move to B
→ New dicovery : $B \rightarrow D$

3) Move to D
→ New discovery : $D \rightarrow G$

4) Move to G (Goal reached)

Final path : A → B → D → G

Step	Location	Knowledge
Start	A	A→B, A→C
Step 1	B	B→D
Step 2	D	D→G
Step 3	G	Goal reached

Explanation

→ Agent learns environment step by step
→ Does not know full map initially
→ updates knowledge while moving

3) Constraint Statisfaction problem

Variables : A, B, C, D (students)

Domain : (9 AM, 10 AM, 11 am)

Constraints : A ≠ 9 AM
B before C
D ≠ A
C ≠ 11 AM

One valid Assignment

Student	Time
A	10 AM
B	9 AM
C	10 AM × (conflict

Correct Valid Assignment

Student	Time
A	10 AM
B	9 AM
C	10 AM

Final Correct Answer

Student	Time
A	10 Am
B	9 Am
C	10 Am ✗ wait - need all constraints

Proper Valid Assignment

Student	Time
A	10 Am
B	9 Am
C	10 Am ✗ again issue

Final Valid Assignment

Student	Time
A	11 AM
B	9 Am
C	10 Am
D	9 Am ✗ (Same as B allowed, but check A condition)